MTH632 - Complex Analysis and Differential Geometry (Quiz 4)

Question # 7 of 10 (Start time: 04:26:24 PM, 05 January 2026)

Complex exponential function is _____.

Select the correct option

○	one to one function
●	not one to one function

MTH632 - Complex Analysis and Differential Geometry (Quiz 4)

Question # 6 of 10 (Start time: 04:26:10 PM, 05 January 2026)

For any complex number z, $\frac{d}{dz}(\cot z) =$ _____.

Select the correct option

●	$-\csc^2 z$
○	$-\sec^2 z$
○	$\tan z \sec z$
○	$-\sec z \tan z$

MTH632 - Complex Analysis and Differential Geometry (Quiz 4)

Question # 3 of 10 (Start time: 04:25:17 PM, 05 January 2026)

For any complex number z, $\sinh z =$ _____.

Select the correct option

○	$\frac{e^{iz}+e^{-iz}}{2}$
○	$\frac{e^{iz}-e^{-iz}}{2i}$
◉	$\frac{e^{z}-e^{-z}}{2}$
○	$\frac{e^{z}+e^{-z}}{2}$

MTH632 - Complex Analysis and Differential Geometry (Quiz 4)

Question # 2 of 10 (Start time: 04:25:04 PM, 05 January 2026)

Period of the function e^z is _____ degree.

Select the correct option

◉	0
○	360
○	180
○	90

MTH632 - Complex Analysis and Differential Geometry (Quiz 4)

Question # 1 of 10 (Start time: 04:24:52 PM, 05 January 2026)

Period of the function Sin z is _____ degree.

Select the correct option

○	0
○	90
◉	360
○	180

MTH632 - Complex Analysis and Differential Geometry (Quiz 4)

Question # 10 of 10 (Start time: 04:24:13 PM, 05 January 2026)

$$\frac{d}{dz}(log_{\alpha} z) = _____, \text{ for } z = r\, e^{i\theta},\ \alpha < \theta < \alpha + 2\pi.$$

Select the correct option

○	1/(1+z)
●	1/z
○	z
○	iz

MTH632 - Complex Analysis and Differential Geometry (Quiz 4)

Question # 9 of 10 (Start time: 04:24:00 PM, 05 January 2026)

We take the derivative of principal branch of z^c because its principal branch is _____ function.

Select the correct option

○	multivalued
◉	single valued

MTH632 - Complex Analysis and Differential Geometry (Quiz 4)

Question # 8 of 10 (Start time: 04:23:46 PM, 05 January 2026)

Which one of the following is the Euler's formula?

Select the correct option

- $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$
- $e^{i\theta} = \cos\theta + i\sin\theta$

MTH632 - Complex Analysis and Differential Geometry (Quiz 4)

Question # 6 of 10 (Start time: 04:23:00 PM, 05 January 2026)

For any complex number z, $\frac{d}{dz}(\sec z) =$ _____.

Select the correct option

- $-\cot z \sec z$
- $\cot z \sec z$
- $\tan z \sec z$
- $-\sec z \tan z$

MTH632 - Complex Analysis and Differential Geometry (Quiz 4)

Question # 5 of 10 (Start time: 04:22:45 PM, 05 January 2026)

For any complex number z, $\frac{d}{dz}\operatorname{csch} z =$ _____.

Select the correct option

○	$\sec h^2 z$
○	$-\sec hz \tanh z$
◉	$-\operatorname{csc} hz \coth z$
○	$-\operatorname{csc} h^2 z$

MTH632 - Complex Analysis and Differential Geometry (Quiz 4)

Question # 4 of 10 (Start time: 04:22:12 PM, 05 January 2026)

For $z = x + i\,y$, we have
$\sin z =$ _____.

Select the correct option

○	$\cos x \sinh y + i \sin x \cosh y$
○	$\sin x \cos y + i \cos x \sin y$
◉	$\sin x \cosh y + i \cos x \sinh y$
○	$\cos x \sin y + i \sin x \cos y$

MTH632 - Complex Analysis and Differential Geometry (Quiz 4)

Question # 3 of 10 (Start time: 04:21:57 PM, 05 January 2026)

For any complex number z, $\frac{d}{dz}(\tan z) =$ _____

Select the correct option

○	$\tan z \sec z$
○	$-\csc^2 z$
○	$-\sec z \tan z$
●	$\sec^2 z$

MTH632 - Complex Analysis and Differential Geometry (Quiz 4)

Question # 2 of 10 (Start time: 04:21:41 PM, 05 January 2026)

The complex exponential function e^z is onto function.

Select the correct option

○	True
●	False

MTH632 - Complex Analysis and Differential Geometry (Quiz 4)

Question # 1 of 10 (Start time: 04:21:26 PM, 05 January 2026)

For any complex number z, $\frac{d}{dz}\cosh z =$ _____.

Select the correct option

- (●) $\sinh z$
- ( ) $\sec h^2 z$
- ( ) $\sin z$
- ( ) $\tanh z$

MTH632 - Complex Analysis and Differential Geometry (Quiz 4)

Question # 10 of 10 (Start time: 04:28:02 PM, 05 January 2026)

For any complex number z, $\frac{d}{dz}(\tan z) =$ _____.

Select the correct option

○	$-\csc^2 z$
○	$\tan z \sec z$
●	$\sec^2 z$
○	$-\sec z \tan z$

MTH632 - Complex Analysis and Differential Geometry (Quiz 4)

Question # 9 of 10 (Start time: 04:27:06 PM, 05 January 2026)

The principal value of i^i is _____.

Select the correct option

◉	0.20788
○	0.10788
○	0.30788
○	0.40788

Question # 8 of 10 (Start time: 04:26:43 PM, 05 January 2026)

For any complex number z, csch $z =$ _____.

Select the correct option

- $\frac{e^z + e^{-z}}{2}$
- $\frac{2}{e^z + e^{-z}}$
- $\frac{e^z - e^{-z}}{2}$
- $\frac{2}{e^z - e^{-z}}$